

15 Asset-liability management

8 primary questions | Oldest to newest

2026 objectives: 4.6.1-4.6.2

7 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

Click a question below to begin.

CA1-P2 April 2005 Q4	13 marks
CA1-P2 September 2005 Q4	27 marks
CA1-P2 April 2006 Q7	23 marks
CA1-P2 September 2006 Q5	15 marks
CA1-P2 April 2008 Q1	7 marks
CA1-P2 April 2012 Q7	20 marks
CA1-P2 April 2014 Q1	8 marks
CA1-P2 September 2018 spare Q7	26 marks

CA1-P2 | April 2005 | Q4

Current topic | 13 marks total | Original PDF p2

Related: Ch 17, Ch 36.

- 4** A general insurance company is constructing an asset-liability model to determine the likely future relationship between its assets and liabilities.
- (i) Describe the features of the business that the model will need to take into account if it is to be used to assess future solvency. [5]
 - (ii) Discuss the issues that should be considered when deciding whether it would be more effective to use a deterministic or a stochastic model for this purpose. [4]
 - (iii) List the parties who may be interested in the level of the solvency of the company, and state a reason for the interest of each. [4]
- [Total 13]

CA1-P2 | September 2005 | Q4

Current topic | 27 marks total | Original PDF p2

Related: Ch 1, Ch 34.

- 4** The government of a certain country has just passed legislation creating a Fund for Protecting Pensions (FPP). When a sponsor of a defined benefit scheme becomes insolvent and the scheme is unable to provide the benefits that have been promised to members, the FPP will take over the scheme in order to guarantee benefits. The FPP will be financed by levies on the sponsors of defined benefit schemes and by the assets of schemes that it takes over.
- (i) List the parties which will be affected by the imposition of levies, and state how each is affected. [6]
- (ii) Discuss the potential conflicts of interest that an actuary might face in advising the FPP on the levies it should charge, and comment on how these potential conflicts could be managed. [5]

CA1-P2 | September 2005 | Q4 - continued[Current topic](#) | [27 marks total](#) | [Original PDF p3](#)Related: [Ch 1](#), [Ch 34](#).

A few years after the FPP has come into effect, all defined benefit schemes in the country have been terminated. Schemes have either been wound up after securing all their liabilities, or their assets and liabilities have been transferred to the FPP. The FPP has assets of around \$10 billion to meet the benefits of schemes it has taken over. The government has made it clear that it will not provide any funds if the FPP were unable to meet all the benefits promised.

- (iii) Discuss the factors the FPP should consider at this point in setting its investment strategy and managing its liabilities. [16]
[Total 27]

CA1-P2 | April 2006 | Q7

Current topic | 23 marks total | Original PDF p3

Related: Ch 16, Ch 3.

- 7** A long-established company operates in a country with a developed economy and developed capital markets. The company sponsors a pension benefit scheme for its employees.
- (i) State the principles for setting the scheme's investment strategy. [2]
 - (ii) Describe how the nature of the scheme's liabilities could influence its choice of assets. [10]
 - (iii) List the controls over pension scheme investment that the country's regulatory authorities might impose. [5]
 - (iv) Describe the process the company could use when assessing the scheme's investment experience. [6]
- [Total 23]

CA1-P2 | September 2006 | Q5

[Current topic](#) | [15 marks total](#) | [Original PDF p3](#)

Related: Ch 26.

- 5 A life insurance company prices level immediate annuity contracts on the assumption that it will earn an investment return 0.75 percentage points per annum in excess of the yield on appropriately dated government bonds. It invests 75% of the single premiums received for this contract in corporate bonds and 25% in ordinary shares.

Discuss whether this investment policy is appropriate, identify the risks to the company from following it, and suggest possible alternative investment strategies.

[15]

CA1-P2 | April 2008 | Q1

Current topic | 7 marks total | Original PDF p2

Related: Ch 14.

- 1**
- (i) State the principles of investment. [2]
 - (ii) Describe the actuarial techniques that can be used to determine an investment strategy. [5]
- [Total 7]

CA1-P2 | April 2012 | Q7

Current topic | 20 marks total | Original PDF p4

Related: Ch 14.

- 7**
- (i) Discuss how a portfolio of assets could be constructed to match liabilities. [4]
 - (ii) Outline why it might be important for an investor to select assets that match their liabilities. [3]
 - (iii) Discuss the extent to which the investments of a defined benefit pension scheme that is closed to new members and future accruals could be mismatched relative to its liabilities. [5]

A couple both aged 50 have four children aged between 10 and 16. They have been investing part of their resources with the aim of providing enough money to pay for university tuition fees for all four children. They have assumed that tuition fees in future will increase from the current rate in line with inflation.

A substantial change to university tuition fees has been announced for the following year. They will increase to three times the current rate.

- (iv) Explain how this increase in fees may affect the couple's future financial planning. [8]
- [Total 20]

CA1-P2 | April 2014 | Q1

Current topic | 8 marks total | Original PDF p2

Examiner report and original paper: use the links for this question in [index.html](#).

- 1** Describe the likely cashflows and appropriate matching investments for the following:
- (a) professional indemnity insurance contracts
 - (b) income protection insurance contracts
 - (c) critical illness insurance contracts
 - (d) a €1m prize which is on offer for proving an unsolved mathematical theorem
- [8]

CA1-P2 | September 2018 spare | Q7

Current topic | 26 marks total | Original PDF p4

2018 spare paper (X18), distinct from the September main sitting. Original paper and matching spare examiner report restored from the supplied archive. Related: Ch 16.

7 An insurance company writes immediate annuities.

- (i) Discuss how a portfolio of assets could be constructed to match the insurance company's liabilities. [6]
- (ii) Outline why exact matching is likely to be difficult to achieve. [3]
- (iii) Explain why matching is likely to be more difficult to achieve for annuities with high escalation rates compared with level annuities. [2]

The company is considering adopting an unmatched investment strategy for its immediate annuity business.

- (iv) Discuss the factors that the company should consider before deciding to adopt an unmatched strategy. [7]

The company decides to adopt an unmatched strategy. Investment managers will be expected to follow an active investment strategy which will be monitored relative to specified benchmarks.

- (v) List reasons why the actual investment returns achieved by the investment managers could be different from the benchmark returns. [4]

An external investment management company is to be chosen to manage the unmatched strategy. Prior to concluding a contract with this investment management company, the insurance company needs to complete a due diligence exercise.

- (vi) Outline the features of the investment management company's practices and procedures which the insurance company should consider as part of this assessment. [4]

[Total 26]